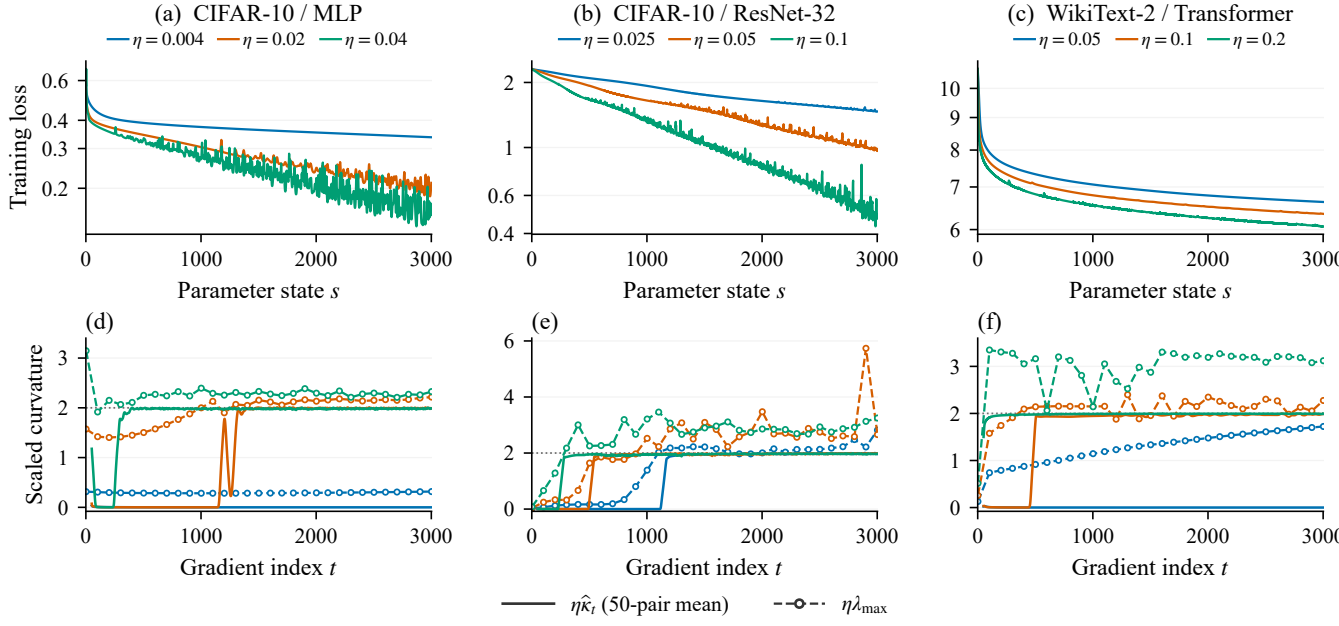

(a) CIFAR-10 / MLP
η = 0.004
η = 0.02
η = 0.04
Training loss
Parameter state s
(b) CIFAR-10 / ResNet-32
η = 0.025
η = 0.05
η = 0.1
Parameter state s
(c) WikiText-2 / Transformer
η = 0.05
η = 0.1
η = 0.2
Parameter state s
(d)
Scaled curvature
Gradient index t
(e)
Gradient index t
(f)
Gradient index t
η κ̂_t (50-pair mean)
η λ_max